C-P: conditional terminal-interface foundations

14 October 2026 — analytical review, no selected replacement node

Scope. The user selected coupled point merging with local completion at the choke. This note proves interface facts conditional on admissible departure histories, not existence or physical correctness of a new service law. The approved H1/T1 predecessor source/PDF are unchanged. No traffic integration, parameter fit, numerical gradient or optimisation is run.

1. Retain H1 inputs and finite terminal histories

Use the accepted road length $L = L_p + b$, initial footprint $[0, \ell]$, total continuous mass M, fixed advice $v > 0$, and initially empty terminal. H1 produces conservative entrance histories F_j for $j \in \{E, A, B\}$. Let $w, k_j > 0$, $H = k_j b$, $\tau = b/v$, $\tau_b = b/w$ and $C(v) = vwk_j/(v+w)$. Histories are zero at negative times. The prospective node must produce absolutely continuous, nondecreasing Y_j satisfying

$$P_j(t) = F_j(t - \tau), \qquad 0 \le Y_j(t) \le P_j(t), \qquad 0 \le \dot{Y}_j(t) \le C(v) \quad \text{a.e.} \tag{1}$$

At an empty eligible queue the flow must also respect available arrivals; this is included in the requirement $Y_j \le P_j$. Define

$$N_j = F_j - Y_j, \qquad q_j = P_j - Y_j \ge 0. \tag{2}$$

N_j is physical stock; q_j is eligible unserved mass. Unlike the old nominal member, (1)–(2) do not require $Y_A = P_A$. They allow $q_A > 0$. Any extra service-memory variable is separate from vehicle mass. Traffic that has crossed the objective boundary is not counted again as queue stock.

2. The conservative handoff bound does not require zero A delay

Retain the sufficient domain

$$M < H, \qquad 0 \le \dot{F}_j \le C(v), \qquad F_j(t) \le M. \tag{3}$$

The accepted empty-link vacancy formula is

$$V_j(t) = \begin{cases} k_j w t - F_j(t), & 0 < t \le \tau_b, \\ Y_j(t - \tau_b) + H - F_j(t), & t > \tau_b. \end{cases} \tag{4}$$

For early times, $F_j(t) \le C(v)t < k_j w t$ because $C(v) < k_j w$. After τ_b, nonnegative departures give

$$V_j(t) \ge H - M > 0. \tag{5}$$

Thus the terminal entrance remains at its capacity under these conditional first-order histories even if A or E departures temporarily stop. Inflow within that capacity therefore does not restrict H1. This reuses the predecessor's conservative proof, which never used $Y_A = P_A$. It is not a general spillback solution or a proof that a microscopic reference fits the chosen storage domain.

3. Complete local TST includes every eligible lane queue

Assume the eventual node clears the cohort in finite time T and there are no later arrivals or lost mass. The measured population remaining before the choke is $M - \sum_j Y_j(t)$. Therefore

$$\mathrm{TST}_{\mathrm{choke}} = \int_0^T \left[M - \sum_j Y_j(t) \right] dt \tag{1}$$

$$= \int_0^T \left[M - \sum_j P_j(t) \right] dt + \int_0^T \sum_j q_j(t) dt \tag{2}$$

$$= \frac{M(L - \ell/2)}{v} + \int_0^T (q_E + q_A + q_B)(t) dt. \tag{6}$$

The last equality uses unchanged H1's uniform initial footprint and common ideal progress speed, its paired lateral conservation, and terminal minimum travel b/v. Lateral exchange changes lane composition, not the sum of minimum-travel completion histories. If B's old nonqueuing law is retained, $q_B = 0$ and the new term is A's queue area. Equation (6) is an accounting identity, not a flow solver or a prediction of optimal advice. Post-choke travel is excluded, while its service influence may remain in the node state/history. Such influence is not added to (6) as fictitious mass.

4. Clearance needs a new service argument

All H1 mass is minimum-travel eligible by $t_* = L/v$, so $\sum_j P_j(t_*) = M$. Afterwards let $Q(t) = \sum_j q_j(t)$. A sufficient property of a future node is

$$\sum_j \dot{Y}_j(t) \geq \mu_{\min} > 0 \quad \text{whenever } Q(t) > 0 \text{ and } t \geq t_*. \tag{7}$$

Then $\dot{Q} = -\sum_j \dot{Y}_j \leq -\mu_{\min}$ while nonempty, and

$$T \leq t_* + \frac{Q(t_*)}{\mu_{\min}}. \tag{8}$$

No such lower bound has been established for a new coupled service law. It is sufficient, not necessary: another valid law-specific clearance proof can replace it. The old R/α guarantee cannot be copied when the receiving/service state has changed. Likewise these interface facts do not inherit a generic junction well-posedness theorem.

5. Conditional compact initial information

Under continuing symmetry the admitted initial information reconstructs

$$a_0 = b_0 = (\rho_0 - e_0)/2. \tag{9}$$

If all physical parameters and the new service initial state/history are shared and known, H1 fixes the same F_j for A+E and G. If the prospective node is a unique autonomous causal map of those histories and its known initial state, both representations give the same Y_j, queue histories and (6). Predicted service memory does not by itself require an extra initial observation. This is a

conditional interface deduction; the new map's existence/uniqueness and the old action/gradient certificate are not proved here. An unknown initial memory would reopen G2.

Source basis. Jin, continuous LTM (2015), selected demand/vacancy and boundary-history sections support the physical interface. All bounds, extended accounting and conditional reconstruction above are deductions for the stated candidate inputs and admissible departures, not source theorems about an unspecified coupled node.